Report Title

Course

Industrial Engineering and Management

Group Number or other stuff that needs to be on the front page

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Course coordinator:
Name of the Professor

Number$^{\mathrm{th}}$ of Month 2019

for an alternative layout

Abstract

Some text that can be replaced to make a nice piece of an abstract here.

Keywords— keyword1, keyword2, keyword3...

Contents

Chapter 1: The basics

1.1 Referring and citing

1.1.1 Citing options

This information came from (Chapallaz et al., 1992) is done using *citep*{}, however, you can also use citing without brackets such as "according to Chapallaz et al. (1992) ... " with *citet*{}

1.1.2 Refer to a single figure

Here we make a reference to figure 1.1. We can also make the work "figure" part of the reference, i.e. Figure 1.1. Chapallaz et al. (1992)

Figure 1.1: Caption of the figure

1.1.3 Refer to the appendix

We have placed additional information in the appendix, it is placed in Appendix A or if you want to refer to the figure specifically, then we can use Figure A.1.

1.2 Equations

In LaTeXyou can make amazing equations quite easily. Especially, if you have Premium Overleaf (for free with your RUG account). At the top of your file press the Ω button, this will open a menu with almost all math symbols, saving you a lot of time remembering the names.

First, a simple equation is shown in Equation 1.1

$$c = \sqrt{a^2 + b^2} \tag{1.1}$$

We can also do this equation inline, such as $c = \sqrt{a^2 + b^2}$. But this is not always possible and may look bad.

In Equation 1.2 we see a number, the first line has no number as we used "*nonumber*", and the last equation is referenced separately as Equation 1.3

$$c = \sqrt{a^2 + b^2}$$

$$e^{i\pi} + 1 = 0 \tag{1.2}$$

$$h_{ij}(t) = \frac{1}{d_{ij}(t)} e^{\left(\frac{-2\pi\sqrt{-1}}{\lambda} d_{ij}(t)\right)} \tag{1.3}$$

You can also make things fancy using colours and arrows, such as shown in Equation 1.4. However, make sure it is readable and doesn't become a circus. Here, we use the package "Tikz" to create the arrows. Effectively, we create an image and place this over our equation. This can become quite difficult to read in the code, so just mess around with things and use the internet.

$$\underbrace{\beta(k)}_{\text{larger}} = ||l^* - p(k) - \underbrace{Tu(k)}_{\text{larger}}||^2 - ||y(k)v(k) - \underbrace{Tu(k)}_{\text{larger}}||^2 \tag{1.4}$$

1.2.1 Brackets

Let's take a look at some options for brackets in formulas with matrices. A lot of brackets can be used, which you can find here `https://www.overleaf.com/learn/latex/Brackets_and_Parentheses`. However, it is often just as easy to create a matrix to obtain the same result. We see an example in Equation 1.5. You can find each matrix bracket type here `https://www.overleaf.com/learn/latex/Matrices`.

$$\mathbf{a}(\theta, \phi) = \begin{bmatrix} e^{\left(\frac{2\pi\rho_k\sqrt{-1}}{\lambda}\right)} \sin(\theta)\sin(\xi_k)\cos(\phi - \varphi_k) + \cos(\xi_k)\cos(\theta) \\ \vdots \\ e^{\left(\frac{2\pi\rho_l\sqrt{-1}}{\lambda}\right)} \sin(\theta)\sin(\xi_l)\cos(\phi - \varphi_l) + \cos(\xi_l)\cos(\theta) \end{bmatrix} \tag{1.5}$$

1.2.2 Leftover mathematical expressions

Here is a summation example in Equation 1.6, an integral in Equation 1.7, some cases in Equation 1.8 and a minimization problem in Equation 1.9.

$$F_{ij}(t) = \left| \sum_{t=t_k}^{t_{k+M}} h_{ij}(t)a(\theta, \phi)(t) \right|^2 \tag{1.6}$$

$$\int_a^b x^2 \, dx \tag{1.7}$$

$$\begin{cases} p(k+1) = p(k) + Tu(k) \\ y(k) = \|l^* - p(k)\| \end{cases} \tag{1.8}$$

$$\begin{aligned} \min_x \quad & f(x) = x^\mathsf{T} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} x - \mathbb{E}[u] \begin{bmatrix} 8 & 16 \end{bmatrix} x \\ \text{s.t.} \quad & x \geq 0 \end{aligned} \tag{1.9}$$

1.2.3 Bonus Tip

Install Mathpix (demo video: `https://mathpix.com/image-to-latex`). You can create LaTeXformulas from screenshots, saving you a lot of time!

Chapter 2: Multiple figures

2.1 Section 1

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Chapallaz et al. (1992)

2.1.1 Referring to the figure

Here we reference figure 2.1a and 2.1b, but we can also reference like this: Figure 2.1.

(a) Left figure

(b) Right figure

Figure 2.1: Two figures next to each other.

However, we can also make multiple figures next to each other and underneath each other using *subfloats*. This syntax might be a bit more confusing, but is shorter to write.

(a) Caption 1, full textwidth

(b) Caption 2, 0.5 textwidth

(c) Caption 3, 0.5 textwidth

(d) Caption 4, width in inches and also a simple equation $c = \sqrt{a^2 + b^2}$

(e) Caption 5, width in centimetres. Actually, we can place as much text as we want, we can even manually use enters

as you see here

Figure 2.2: The overall caption for the five figures shown. Make sure to make this as elaborate as possible. Without any context outside of the figure, people should understand the figure.

Chapter 3: Code in LaTeX

3.1 Section 1

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Chapallaz et al. (1992).

3.1.1 lstlisting to input code

text, reference Code 3.1 and Code 3.2. Note it will be referred to as "Code x". This is because we have renamed this in the Rest/Paper.sty document. Normally we would get "Listing x". If you prefer this remove "" from the Rest/Paper.sty file. Chapallaz et al. (1992).

Code 3.1: code example for Matlab

```
function res = auto_correlation(y, n)
    N = length(y)-n;
    y = y - mean(y);
    temp = y(end-N+1:end)'*y(end-N-n+1:end-n);
    res = temp/N;
end
```

Code 3.2: code example for Python

```
def foo(n): #recursive function to determine a Fibonacci number
    fib = 0
    if n ≥ 2:
        n_min_one = foo(n-1)
        n_min_two = foo(n-2)
    else:
        n_min_one = 1
        n_min_two = 0
    fib = n_min_one + n_min_two
    return fib #return Fibonacci number

fib_list = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
for item in fib_list:
    print(foo(fib_list))
```

Here we also have a list of our codes. But this will automatically go to the next page.

List of Listings

3.1.2 minted to input code

We can also use “minted” to input our code. Everything works basically the same, it just looks different. However, using minted like this, we do not have the option to assign a label and refer to our code as before.

minted code example with matlab

```matlab
function res = auto_correlation(y, n)
    N = length(y)-n;
    y = y - mean(y);
    temp = y(end-N+1:end)'*y(end-N-n+1:end-n);
    res = temp/N;
end
```

minted code example with python

```python
def foo(n): #recursive function to determine a Fibonacci number
    fib = 0
    if n >= 2:
        n_min_one = foo(n-1)
        n_min_two = foo(n-2)
    else:
        n_min_one = 1
        n_min_two = 0
    fib = n_min_one + n_min_two
    return fib #return Fibonacci number

fib_list = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
for item in fib_list:
    print(foo(fib_list))
```

If you do want to refer to the code using Minted, you have to put the code in this file directly (instead of in the Code folder). Now we can refer to 1, however, note now we cannot use the “autoref” function, we only obtain the number.

3.1.3 Conclusion

Many options exist to insert code using LaTeX, choose whichever one you like or find easiest to use.

```python
def foo(n): #recursive function to determine a Fibonacci number
    fib = 0
    if n >= 2:
        n_min_one = foo(n-1)
        n_min_two = foo(n-2)
    else:
        n_min_one = 1
        n_min_two = 0
    fib = n_min_one + n_min_two
    return fib #return Fibonacci number
```

Listing 1: Another example of python code with minted

Chapter 4: Tables in LaTeX

4.1 Link for easier tables

We can make beautiful tables, however making tables directly in LaTeXcan be very annoying. So we use an online tool. I prefer `https://www.latex-tables.com/`. But search on Google for "Latex table generator" and you can find many more.

4.2 Example table

In Table 4.1 we have used the **Booktabs** style, which places a bold horizontal line at the top and bottom of the table.

*Table 4.1: This is a table caption. It is supposed to be placed **above** the table, instead of underneath like for figures.*

	estimated	**true**	**error**
	degrees	degrees	degrees
forward 1m	149.91	150.44	0.53
right 1m	166.96	160.45	-6.51
circle 0.10m radius	154.93	153.99	-0.96
circle 0.30m radius	153.93	153.91	-0.02
circle 0.50m radius	153.93	153.93	0.00
circle 0.70m radius	138.89	153.89	15.00
circle 0.90m radius	145.91	153.86	7.95

If you type "\begintable", you will obtain a simple template. Some placeholder values have been placed in Table 4.2 for this example. Do note, the template overleaf gives places the caption below the table, so change this yourself! You can also change the placement of the table using [h], [H], etc. the same as for figures, or use [!htb] which works very well in almost al cases.

Table 4.2: This table contains 4 numbers.

1	2
3	4

Conclusion

Here we give an example of how to create sections without a number, but still add them to the table of contents.

Section 1

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Subsection 1

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Bibliography

Chapallaz, J.-M., Fischer, G., and Eichenberger, P. (1992). Manual on pumps used as turbines. Last accessed 10 January 2020.

Appendices

Appendix A: Title of appendix A

You can treat this file as a normal chapter. Just place any text, figures or tables in. The chapter is created in the file /Rest/4_Appendix.tex, but you can add sections, subsections and so on here.

Figure A.1: Caption